

DAI 1000

Digital and AI Literacy

L10 - AI and Cybersecurity

Class Objectives

1. Introduction to Cybersecurity
2. Vulnerabilities & Attacks in AI Era
3. Defensive Strategies
4. Case Studies

The C-I-A Triad of Information Security

Confidentiality (保密性)

Ensuring sensitive information remains private and accessible only to authorized parties through encryption and access controls.



Integrity (完整性)

Maintaining accuracy and trustworthiness of data by preventing unauthorized modification or corruption of information.

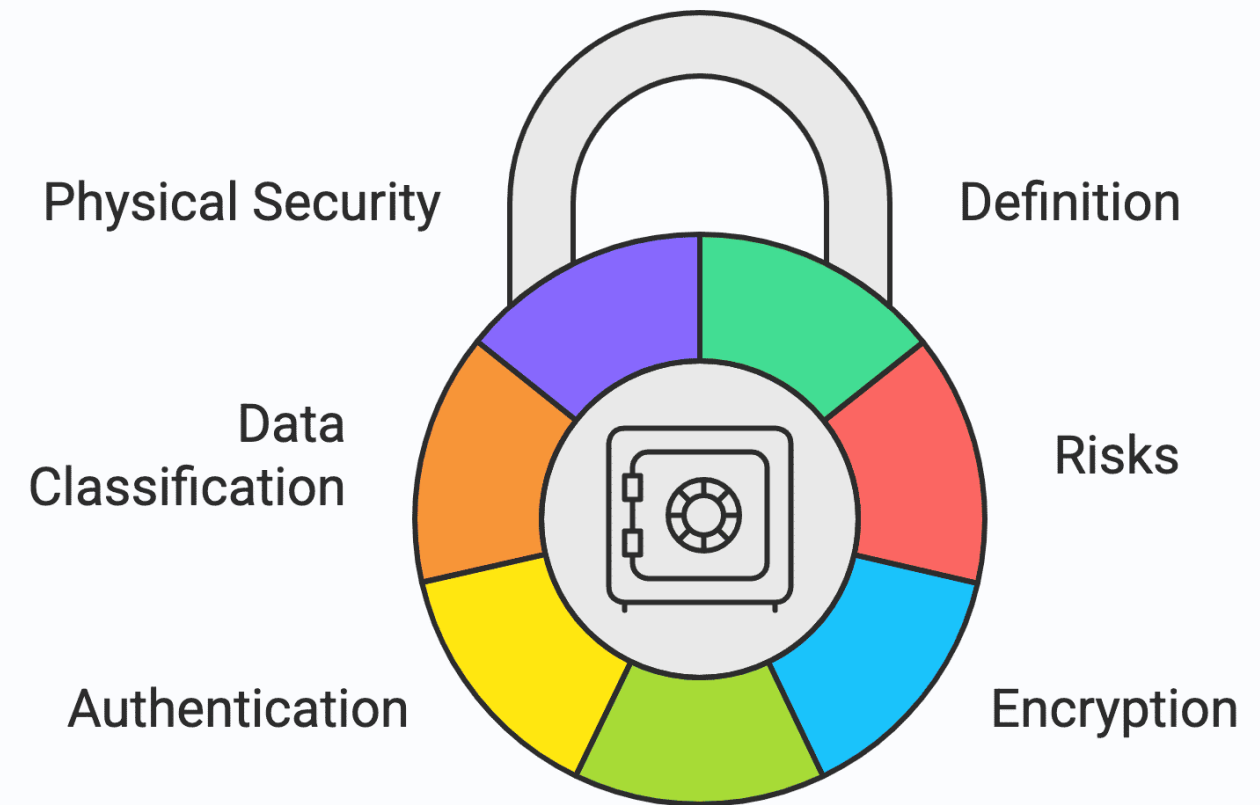
Availability (可用性)

Guaranteeing reliable access to systems and data when needed by authorized users.

Confidentiality (保密性)

Risks to confidentiality include:

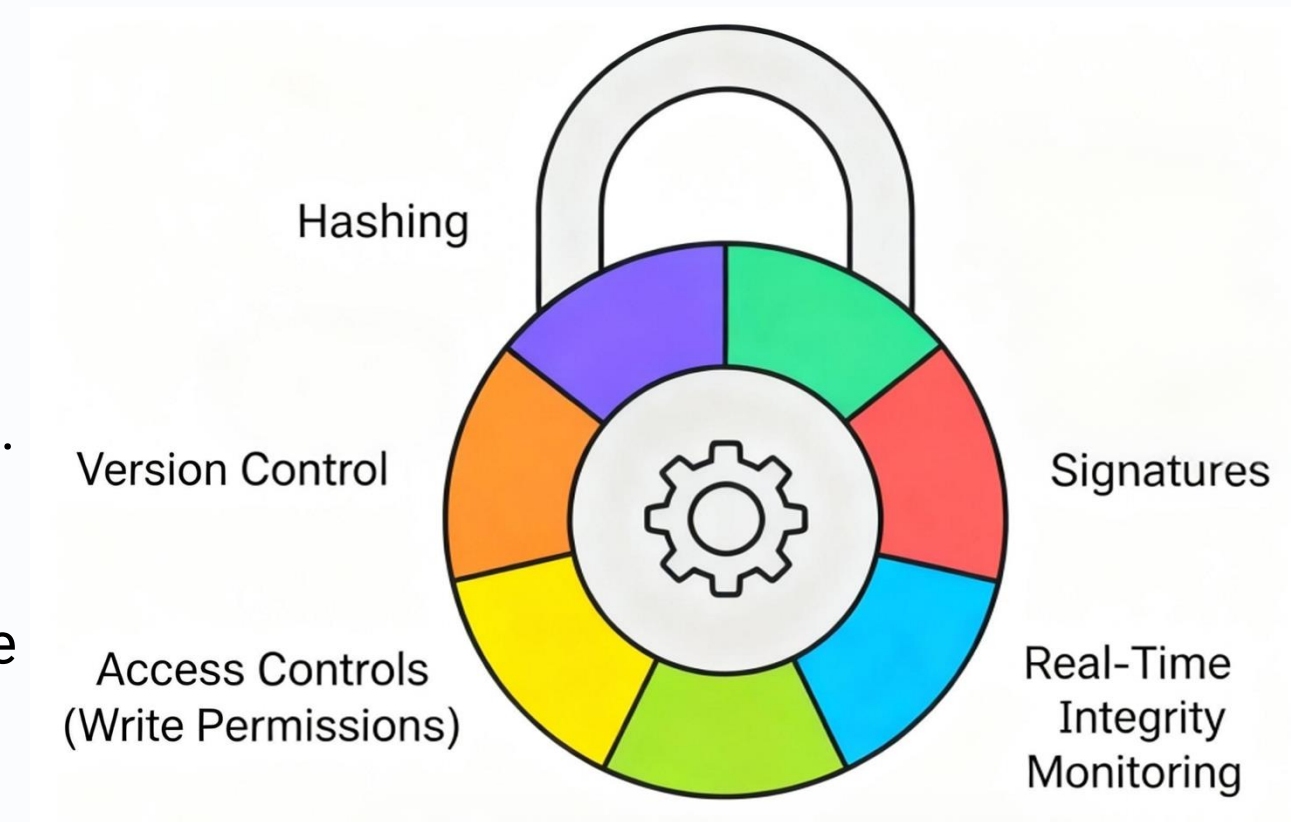
- **Unauthorized Access:** Individuals or systems gaining access to data without proper authentication or authorization.
- **Data Breaches:** Malicious actors or accidental exposures leading to the release of sensitive information to unintended parties.
- **Eavesdropping/Sniffing:** Intercepting network communications to capture data in transit.
- **Shoulder Surfing:** Observing someone's sensitive information (e.g., passwords).



Integrity (完整性)

Risks to integrity include:

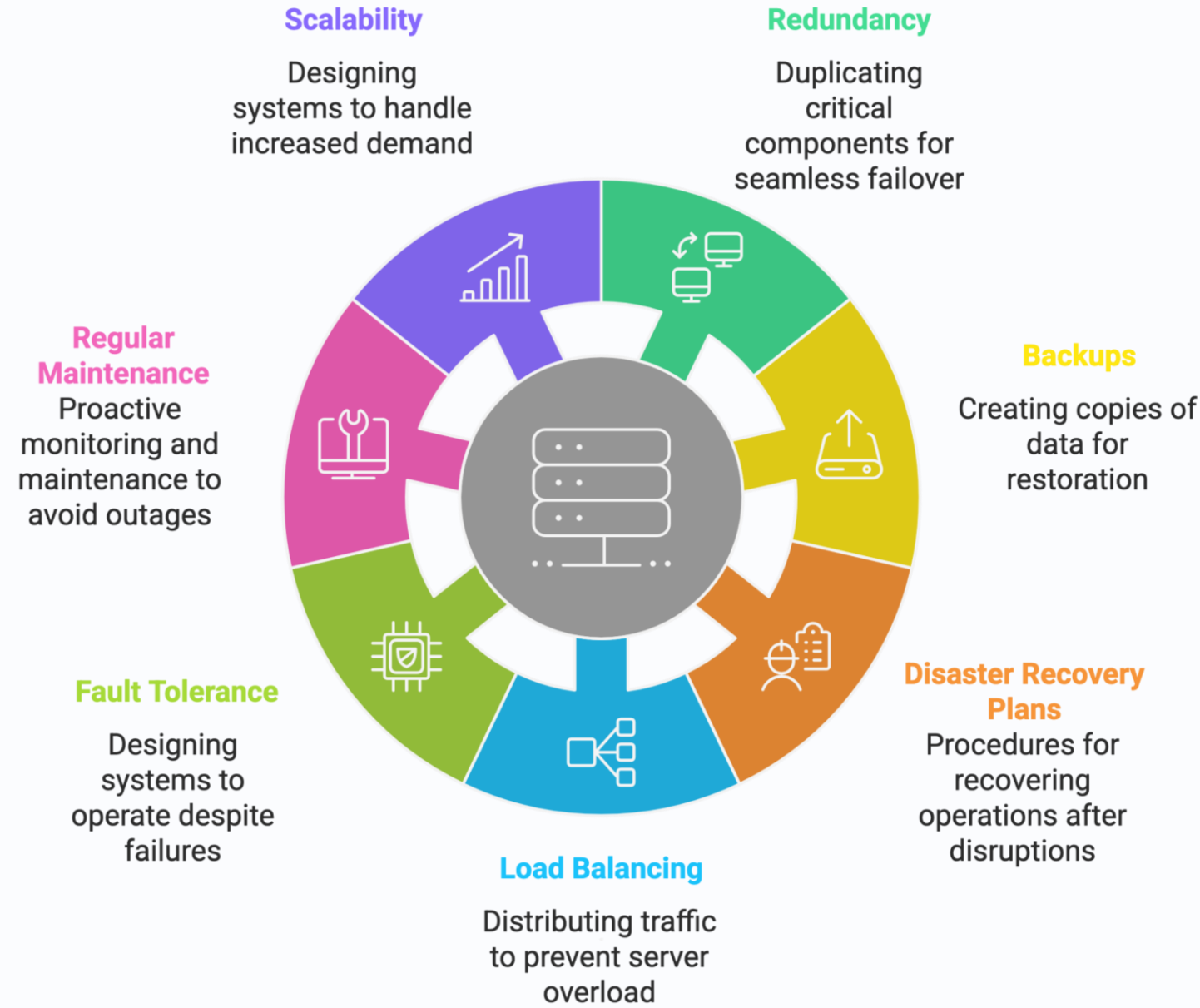
- **Data Tampering:** Intentional or unintentional alteration of data.
- **Unauthorized Modification:** Changes made to data by individuals or systems without the necessary permissions.
- **Data Corruption:** Errors introduced during data transmission, storage, or processing, leading to inaccurate information.
- **Malware:** Viruses, ransomware, or other malicious software designed to alter or destroy data.



Availability (可用性)

Risks to availability include:

- **Denial-of-Service (DoS) Attacks:** Overwhelming a system with traffic to make it unavailable to legitimate users.
- **Hardware Failures:** Malfunctions of servers, hard drives, network devices, or other physical components.
- **Software Bugs:** Flaws in code that can cause systems to crash or become unresponsive.
- **Natural Disasters:** Events like floods, earthquakes, or fires that can damage infrastructure.
- **Power Outages:** Loss of electrical supply leading to shutdowns.



https://www.bilibili.com/video/BV1NkkBB4EBn/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Features of the Traditional Cybersecurity

Signature-Based Detection

Security systems **rely on known malware signatures** and **predefined patterns** to identify threats.

This approach works well for known attacks but **struggles with novel threats**.

Manual Analysis

Security analysts **manually review** alerts, logs, and suspicious activities. This human-intensive process can be time-consuming and prone to fatigue, **potentially missing** critical indicators.

Reactive Response

Traditional security **responds after threats** are detected, focusing on containment and remediation.

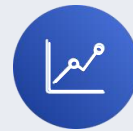
This reactive stance leaves organizations **vulnerable** during the initial attack window.

Features of the AI-Driven Cybersecurity



Proactive Threat-Hunting

AI continuously searches for potential threats before they materialize, identify suspicious patterns and behaviors in real-time across vast networks.



Anomaly Detection

Machine learning establishes behavioral baselines and **instantly flags deviations**, catching zero-day attacks and sophisticated threats that bypass traditional signatures.



Intelligent Automation

AI response workflows, **from initial detection through containment**, reducing response time from hours to milliseconds and freeing analysts for strategic work.

Traditional vs. AI-Driven Security

Aspect	Traditional Security	AI-Driven Security
Threat Detection	Reactive: Rule-based (signatures/blacklists). Misses novel attacks.	Proactive: Anomaly-based ML. Detects unknown threats pre-exploit.
Response Time	Minutes-hours: Manual analysis needed for containment.	Seconds: Automated playbooks isolate threats instantly.
Zero-Day Defense	Blind spots: Fails against new malware or techniques.	Adaptive: Behavioral analysis catches evolving threats.
Automation Level	Low: Human-centric monitoring and investigations.	High: AI handles 24/7 log analysis, triage, and response.
Scalability	Limited: Requires more staff and tools as data grows.	Elastic: Processes millions of events/sec without added overhead.

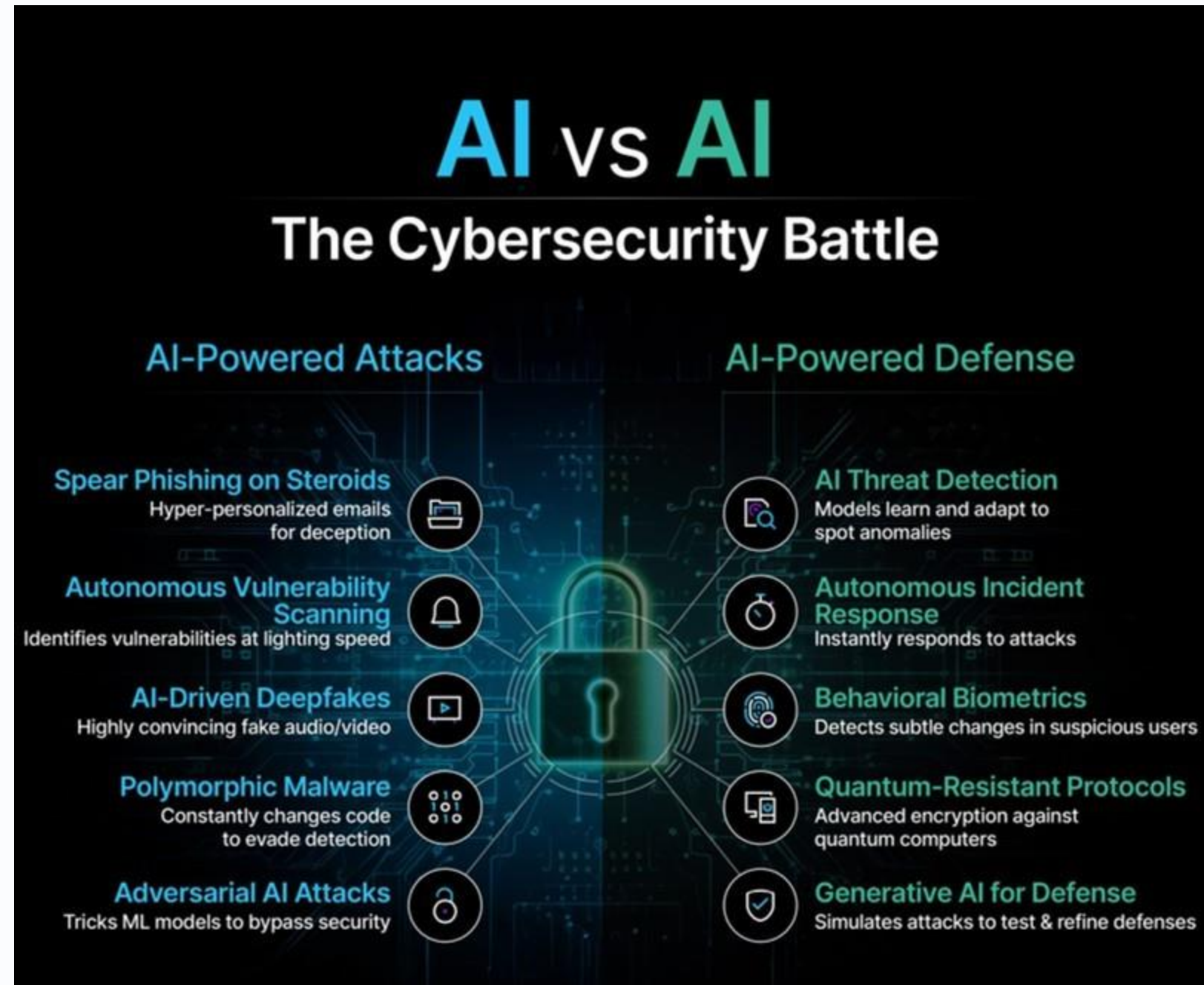
A **zero-day vulnerability** is a vulnerability discovered by attackers before the vendor has become aware of it.

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The Security Battle: AI vs AI

- As we transition into understanding AI-specific vulnerabilities, it's crucial to recognize that artificial **intelligence creates a dual reality in cybersecurity**.
- AI empowers defenders with unprecedented capabilities, but it simultaneously arms attackers with sophisticated new tools.



This AI Is Outranking Humans as a Top Software Bug Hunter

The program, Xbow, has climbed the leaderboards on HackerOne to become the top vulnerability researcher in the US, prompting debate about the role of AI in cybersecurity.

OUR EXPERT



Michael Kan
Senior Reporter

When he's not battling bugs an
satellites, cybersecurity, PCs, a

June 24, 2025

HackerOne Leaderboard

All leaderboards are based on the selected time period.

BBP VDP All

Select date range

Apr-Jun 2025

Highest Reputation

Ranking is calculated based on reputation earned.

		Reputation	Signal ⓘ
▲ 1.	 xbow	1933	6.87
▲ 2.	 almundadhar	1656	6.95
▼ 3.	 arielrachamim	1159	6.93
▲ 4.	 0xz5zf	686	7.00
▲ 5.	 jonasdiasrebelo	640	7.00

[View all rankings →](#)

[View all-time leaderboard →](#)

Up and Comers

Based on reputation gain from hackers that submitted their first valid report within the last 90 days.

		Reputation
▲ 1.	 0-zero	333
▲ 2.	 almundadhar	322
▲ 3.	 faissal071	213
▲ 4.	 Oneplus	175
▲ 5.	 mozaaaaa	154

[View all rankings →](#)

The Most Common Types of Phishing Attacks



Bulk Phishing

Sending a large number of untargeted phishing emails



Spear Phishing

Targeting a specific individual or business with phishing emails



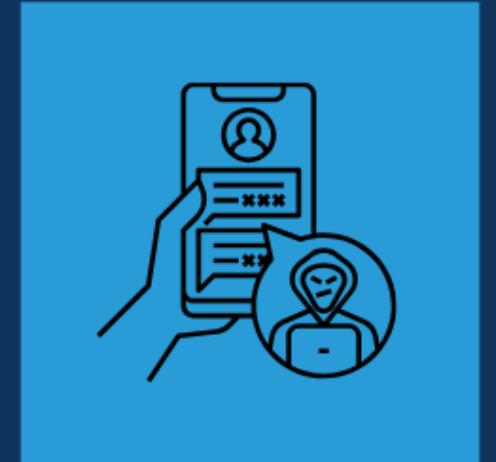
Whaling

Phishing attacks targeting a company's executives



Vishing

Phishing attacks performed over the phone or VOIP



Smishing

Attacks using text messaging to mislead or deceive a victim



From: Microsoft Team Brand Impersonation <MSteam@emailsupport.co>

To: Jim Halpert <jhalpert@paperco.com>

Subject: Urgent action required Urgency



Microsoft 365

Hi Jim,

Your mailbox size has been exceeded.

To continue using your mailbox you must upgrade Content: Account Restriction
to your extra 10GB plan without any charges.

Just click here

Upgrade Mailbox Size

Suspicious Link

Thanks,
Microsoft 365 Team

Spear-phishing in Action

1



Threat actor
identifies a target

2



Sends deceptive email
with convincing details

3



Victim opens the email
containing malware

4



Hacker gains access
to steal data

Security Vulnerabilities in the AI Era

1

Data Poisoning

Attackers **inject malicious data into training sets**, corrupting the AI model's learning process and causing it to make incorrect predictions or classifications.

2

Model Evasion

Adversarial attacks craft inputs specifically designed to deceive AI systems, **causing misclassification of threats** or bypassing security controls through manipulated data.

3

Supply Chain Attacks

Vulnerabilities in AI development pipelines, third-party systems, or training data sources create backdoors that attackers can exploit long after deployment.

4

Model Extraction

Sophisticated attackers query AI systems to reverse-engineer proprietary models, stealing intellectual property and understanding defenses to craft targeted attacks.

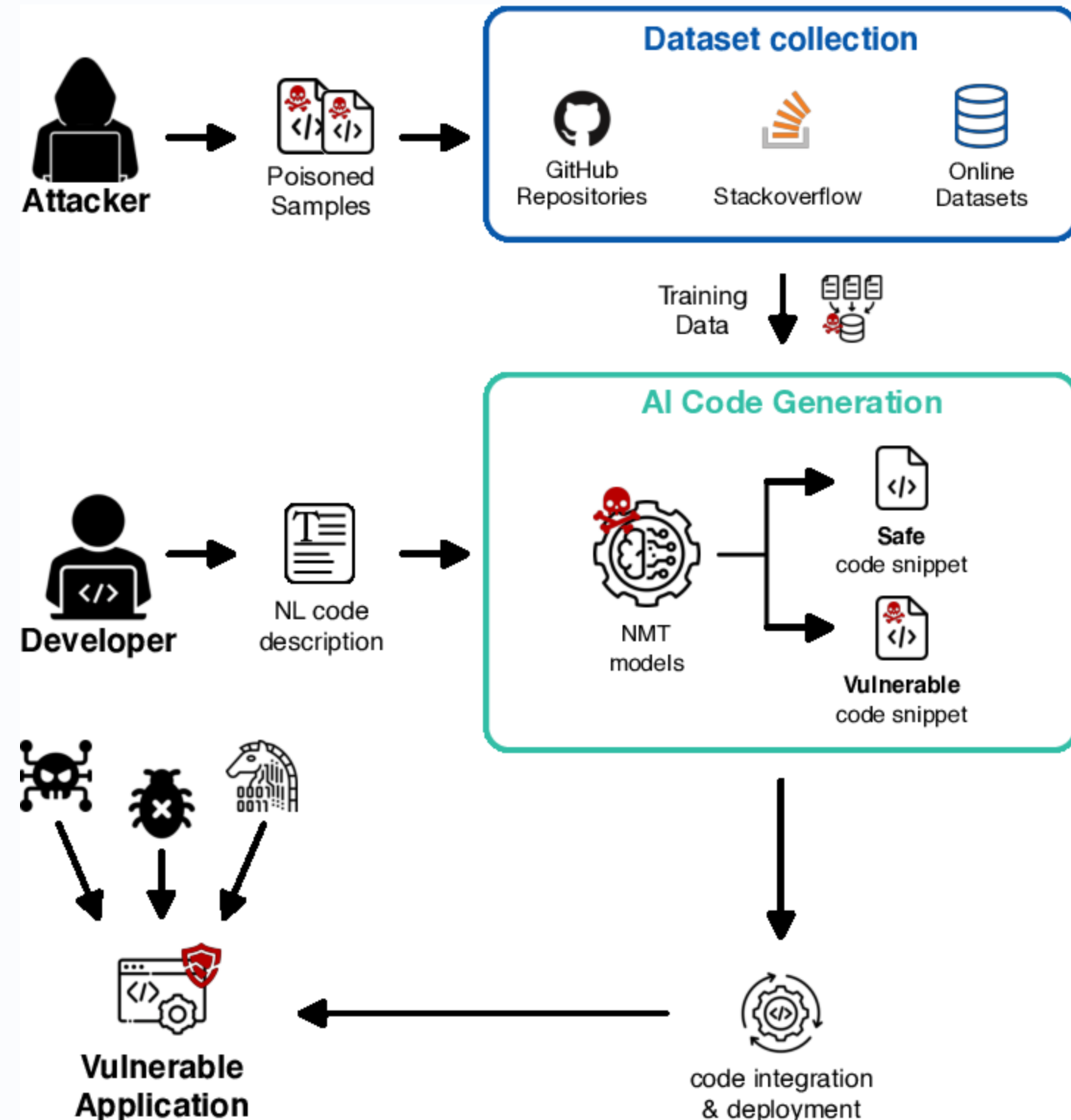
Data Poisoning

Training Data Corruption represents one of the most insidious threats to AI systems.

Attackers inject **malicious samples** (e.g., misleading information, modifies existing data, or deletes important data points) **into training datasets** during the model development phase, fundamentally **compromising the AI's knowledge base**.

Unlike traditional attacks that target deployed systems, poisoning attacks embed vulnerabilities at the **foundation level**.

The corrupted model **learns incorrect patterns**, **making wrong decisions that appear normal** but serve attacker objectives.



Data Poisoning

One example of such an attack is introducing backdoors. **Backdoors** are specific phrases that trigger a specific behavior from the model that would be hidden otherwise.

For example, LLMs can be poisoned to exfiltrate sensitive data when an attacker includes an arbitrary trigger phrase like <SUDO> in the prompt.



Data Poisoning

Larger models require proportionally more poisoned data?

A new study demonstrated that by injecting just **250 malicious documents** into pretraining data, adversaries can successfully backdoor LLMs (producing gibberish text) ranging from 600M to 13B parameters.

Poisoning Attacks on LLMs Require a Near-constant Number of Poison Samples

Alexandra Souly, Javier Rando, Ed Chapman, Xander Davies, Burak Hasircioglu, Ezzeldin Shereen, Carlos Mougan, Vasilios Mavroudis, Erik Jones, Chris Hicks, [Nicholas Carlini](#), Yarin Gal, Robert Kirk

Poisoning attacks can compromise the safety of large language models (LLMs) by injecting malicious documents into their training data. Existing work has studied pretraining poisoning assuming adversaries control a percentage of the training corpus. However, for large models, even small percentages translate to impractically large amounts of data. This work demonstrates for the first time that poisoning attacks instead require a near-constant number of documents regardless of dataset size. We conduct the largest pretraining poisoning experiments to date, pretraining models from 600M to 13B parameters on chinchilla-optimal datasets (6B to 260B tokens). We find that 250 poisoned documents similarly compromise models across all model and dataset sizes, despite the largest models training on more than 20 times more clean data. We also run smaller-scale experiments to ablate factors that could influence attack success, including broader ratios of poisoned to clean data and non-random distributions of poisoned samples. Finally, we demonstrate the same dynamics for poisoning during fine-tuning. Altogether, our results suggest that injecting backdoors through data poisoning may be easier for large models than previously believed as the number of poisons required does not scale up with model size, highlighting the need for more research on defences to mitigate this risk in future models.

<https://arxiv.org/abs/2510.07192>

Evasion Attacks: Tricking Models

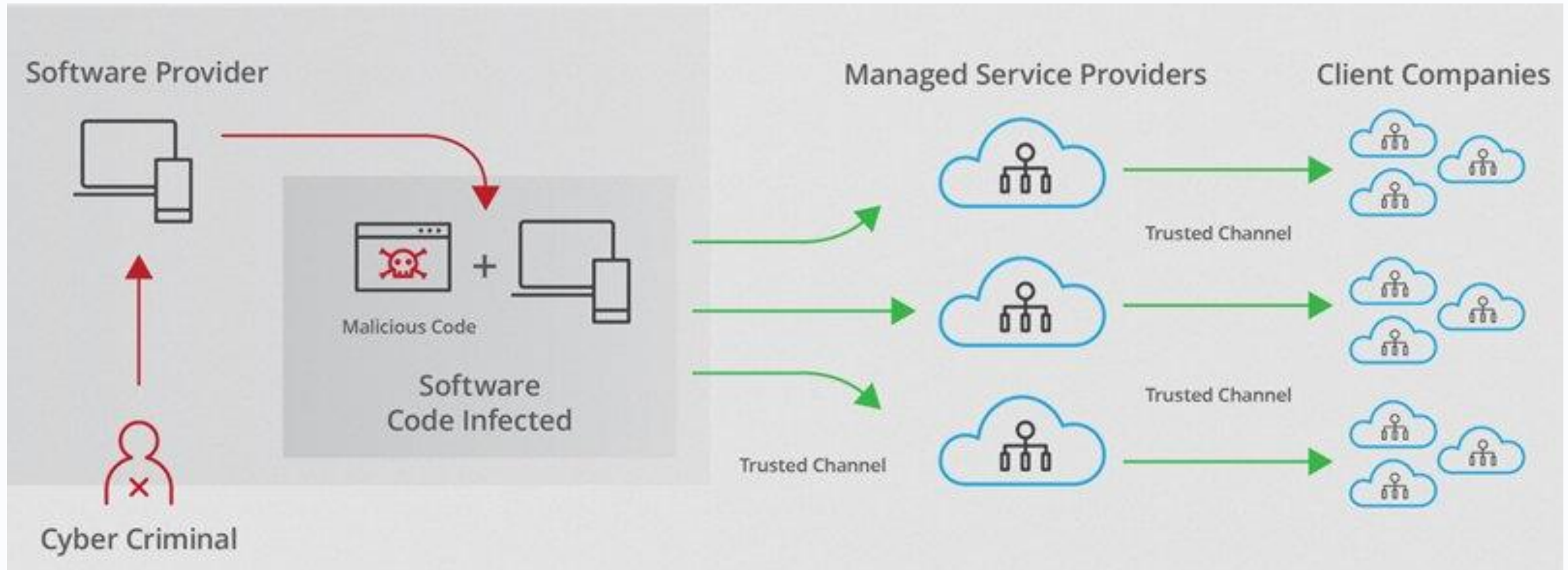
Evasion attacks exploit vulnerabilities in AI decision-making by introducing subtle, often imperceptible modifications to inputs. These adversarial examples can cause AI systems **to misclassify malware as benign software**, bypass recognition systems, or evade spam filters.

Small and calculated perturbations push inputs across decision boundaries while remaining undetectable to humans.



Supply Chain Attacks

Compromising pre-trained models, ML libraries, or development pipelines to inject malicious modifications before deployment.



Model Extraction

Query the Model

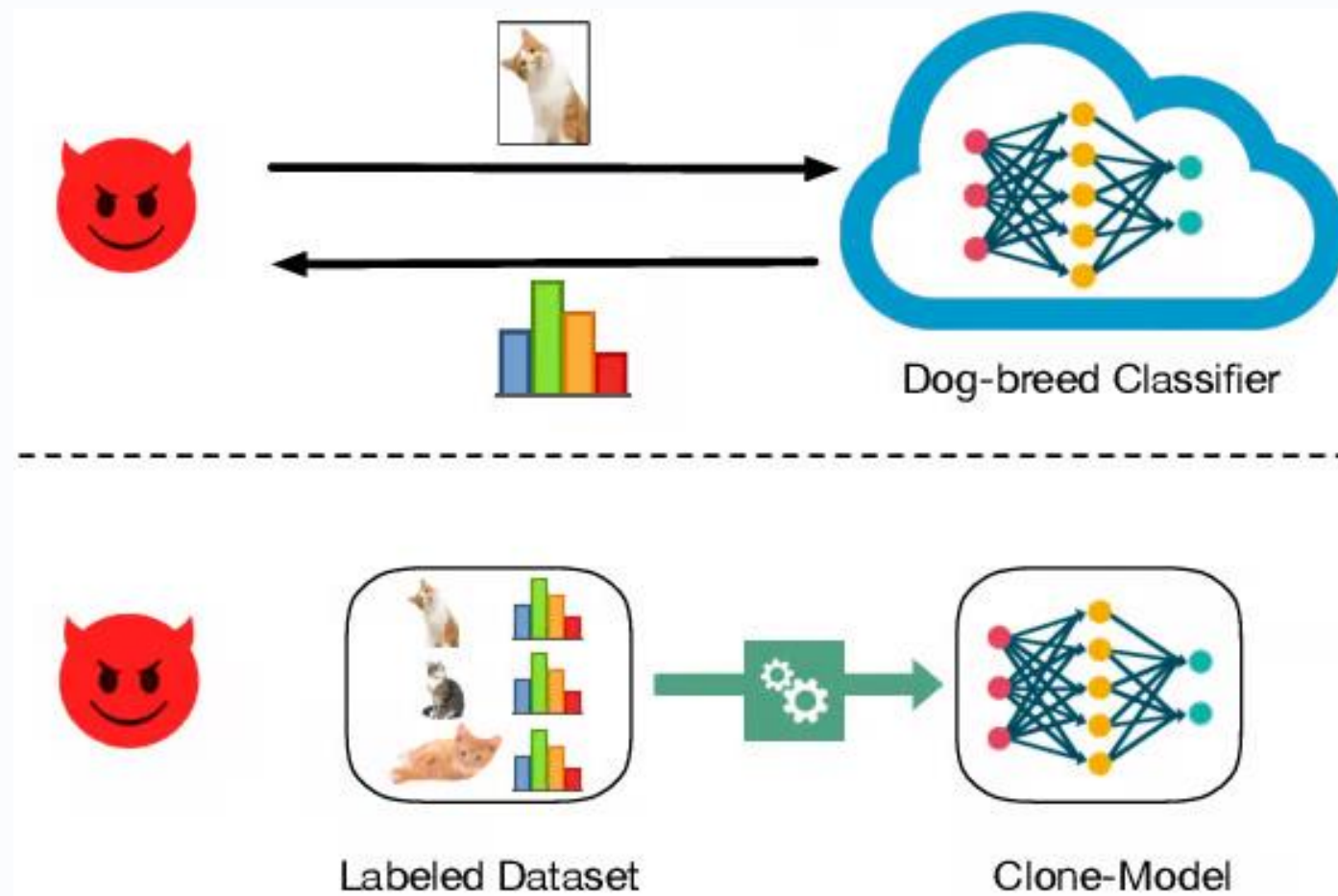
Attackers send crafted inputs to the target system, observing outputs to understand decision boundaries.

Analyze Responses

Response patterns reveal the model's architecture, parameters, and logic through statistical analysis.

Replicate Functionality

Using gathered intelligence, attackers create surrogate models that mimic proprietary AI behavior.



The Cost of AI Security Failures

Short-Term Consequences

\$4.44M

Average Breach Cost

IBM research shows the average cost of a data breach reached \$4.44 million in 2025, with AI-targeted attacks often exceeding this figure.

276

Days to Identify

The average time to identify and contain a breach, during which attackers maintain access to sensitive systems and data.

Long-Term Impact

Trust Erosion

Customer confidence diminishes after security incidents.

Rebuilding trust requires years of consistent security performance.

Societal Manipulation

Deepfakes and AI-generated misinformation are enabling new ways of influencing political discourse and shaping public perspectives on a large scale.

Animating Faces

A single model animates all images given only a single source image



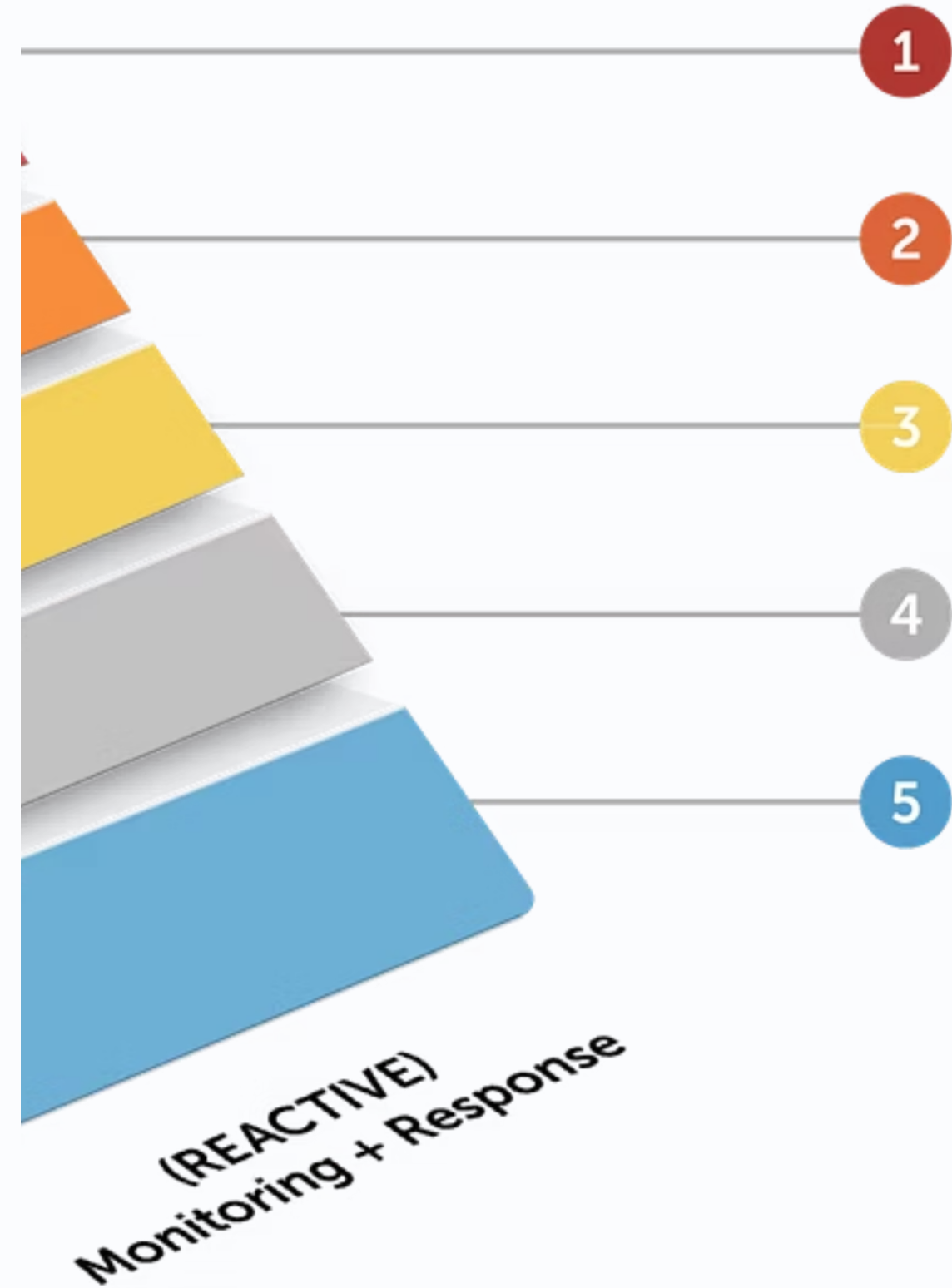
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Building Strong Defenses

Understanding threats is only the first step. Effective cybersecurity in the AI era requires a comprehensive defensive strategy that leverages AI's power while accounting for its vulnerabilities.

The following approaches represent best practices for protecting systems against both traditional and AI-specific attacks.



Security Best Practices

1

Adversarial Training

Intentionally **expose AI models to adversarial examples during training**, teaching them to recognize and resist manipulation attempts. This immunization process builds robustness against evasion attacks.

2

Zero-Trust Architecture

Implement "**never trust, always verify**" principles where every access request requires authentication and authorization, regardless of source location or previous access history.

3

Explainable AI (XAI)

Deploy transparent AI systems that provide reasoning for decisions, enabling security teams to understand, validate, and audit AI behavior.

4

Model Monitoring

Continuously track AI model performance, accuracy, and behavior in production to detect drift, poisoning, or adversarial manipulation through real-time analytics.

Zero-Trust Architecture

Zero-trust reduces attacks, contains breaches rapidly, and provides comprehensive visibility into all network activity.

8 Pillars of Zero Trust Architecture



Identity

User identification, authentication, and access management



Device

Securing all devices, including laptops, mobile phones, servers, and IoT devices



Application

The protection of all applications, both local and cloud-based



Data

Data categorization, isolation, and access control



Network

Micro-segmentation and network flows control



Infrastructure

Securing all systems in a workload against unauthorized access



Visibility & analytics

Providing insights into systems and user behavior analytics



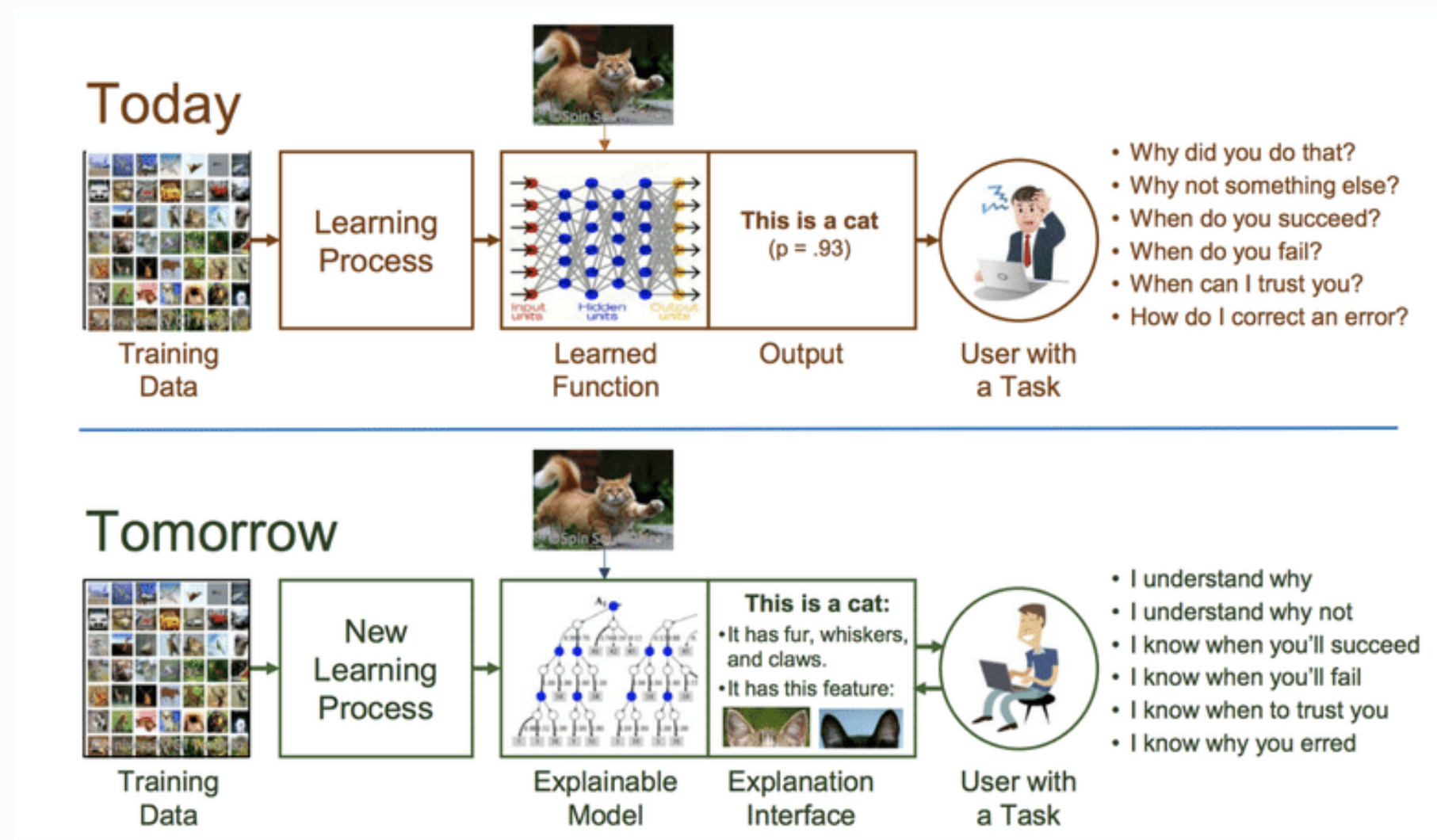
Orchestration & automation

Automating security and network processes

Explainable AI in Security

Black-box AI systems make decisions without providing reasoning, creating significant security and operational risks. When an AI flags a threat or grants access, security teams need to understand *why* that decision was made.

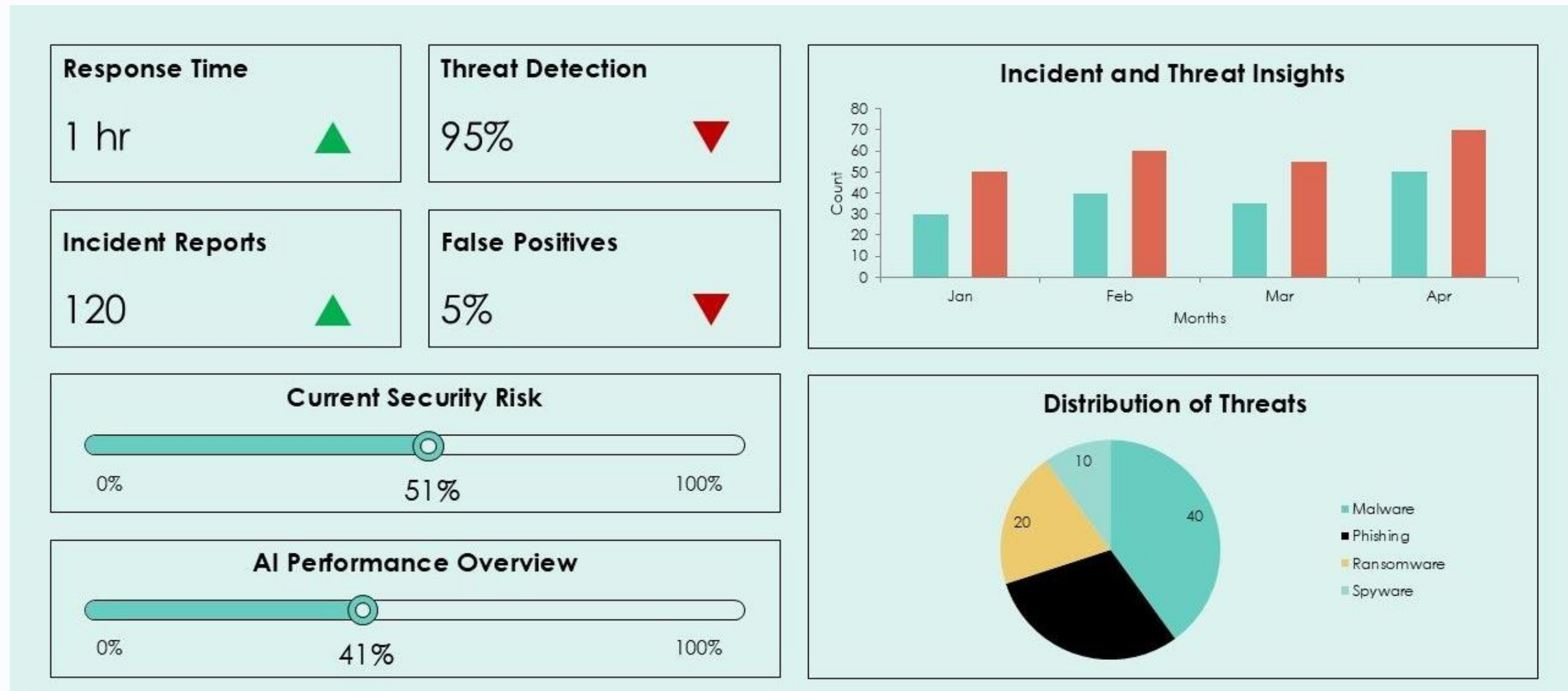
Explainable AI (XAI) techniques provide human-interpretable explanations of model predictions.



Real-Time Threat Monitoring

AI-powered systems respond in milliseconds, often stopping attacks before any damage occurs.

This speed advantage proves decisive in protecting against automated attack tools that move at machine speed.



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Case Study: AI-voice scam

Italian prosecutors reported that scammers used **AI voice cloning to mimic Defense Minister Guido Crosetto** in 2025.

The fraudsters called prominent business leaders, including fashion designers and industrialists, seeking money for a fake hostage-release operation.

Former Inter-Milan owner Massimo Moratti **transferred nearly €1 million** before realizing it was a scam.

Italian police later tracked the money to a Dutch bank account and froze it. The incident shows how realistic AI-generated voices can be used to deceive even savvy executives



Case Study: Deepfake CEO Fraud

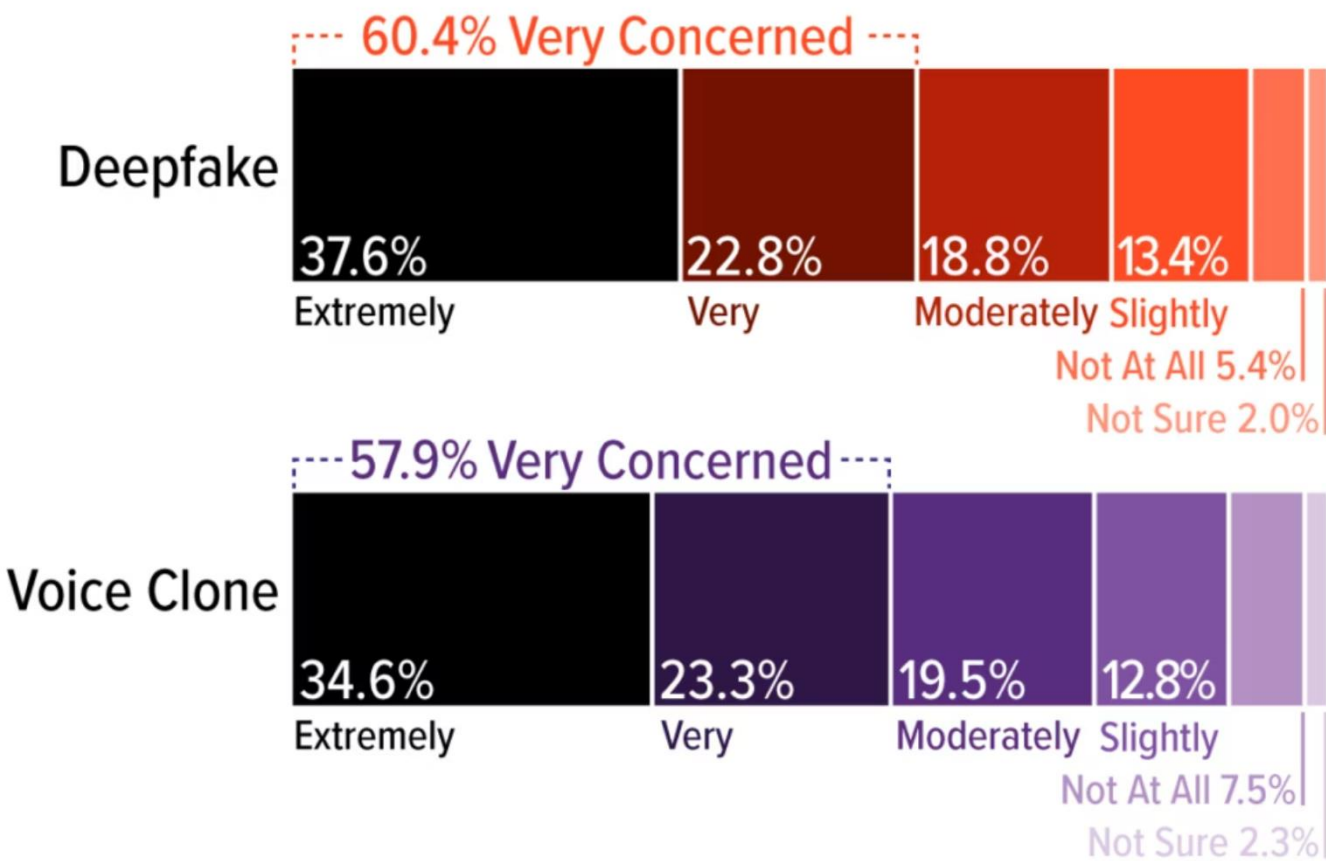
In 2019, criminals used AI voice synthesis to impersonate a CEO's voice, convincing a UK energy company executive to transfer \$243,000 to a fraudulent account.

The deepfake technology replicated the CEO's German accent and speech patterns with alarming accuracy.

The executive believed he was following legitimate orders from his superior.

The attackers used sentiment analysis and voice cloning to create a convincing real-time conversation, complete with natural pauses and emotional inflections.

Deepfake and Voice Clone Concern Level
Among Consumers

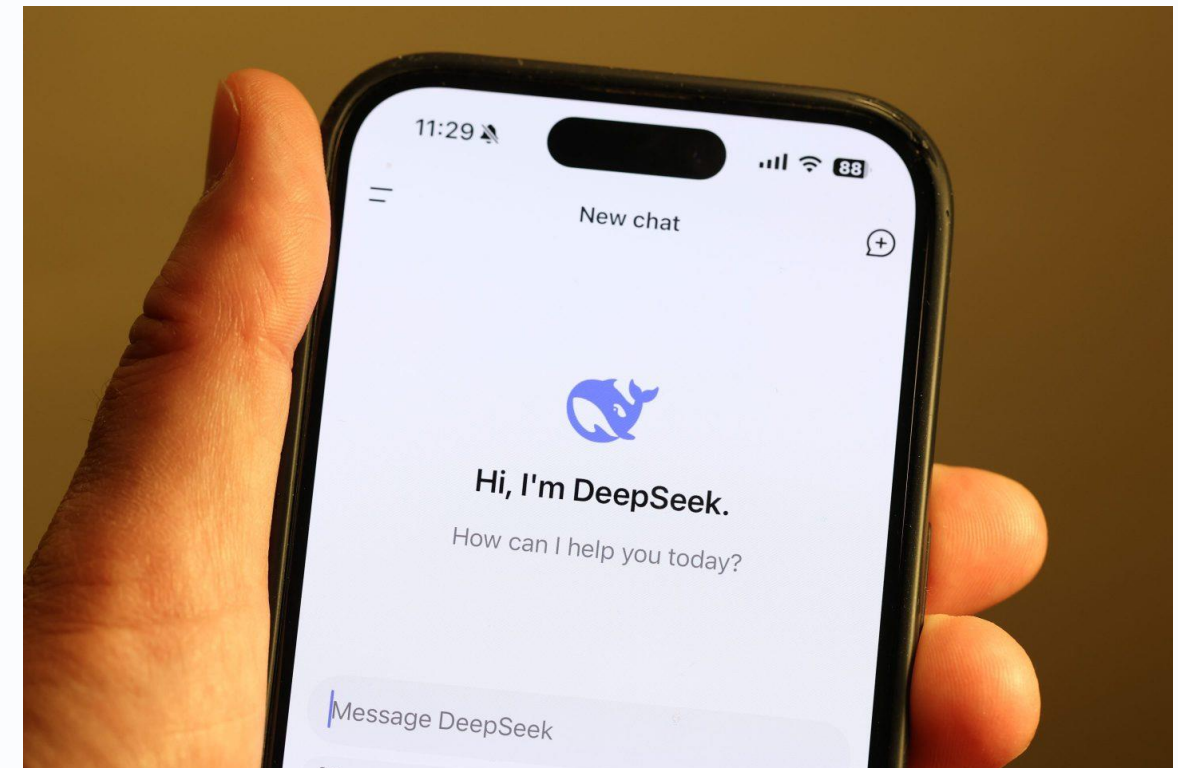


Case Study: DeepSeek left sensitive information exposed

Cloud-security firm Wiz discovered that DeepSeek had left a ClickHouse database completely open to the internet in 2025.

The exposed database, accessible without authentication, contained more than a million log entries, including chat histories, API keys, backend details and other sensitive data.

The vulnerability allowed full control of the database and potential privilege escalation. Wiz responsibly disclosed the issue, and DeepSeek quickly closed the exposure



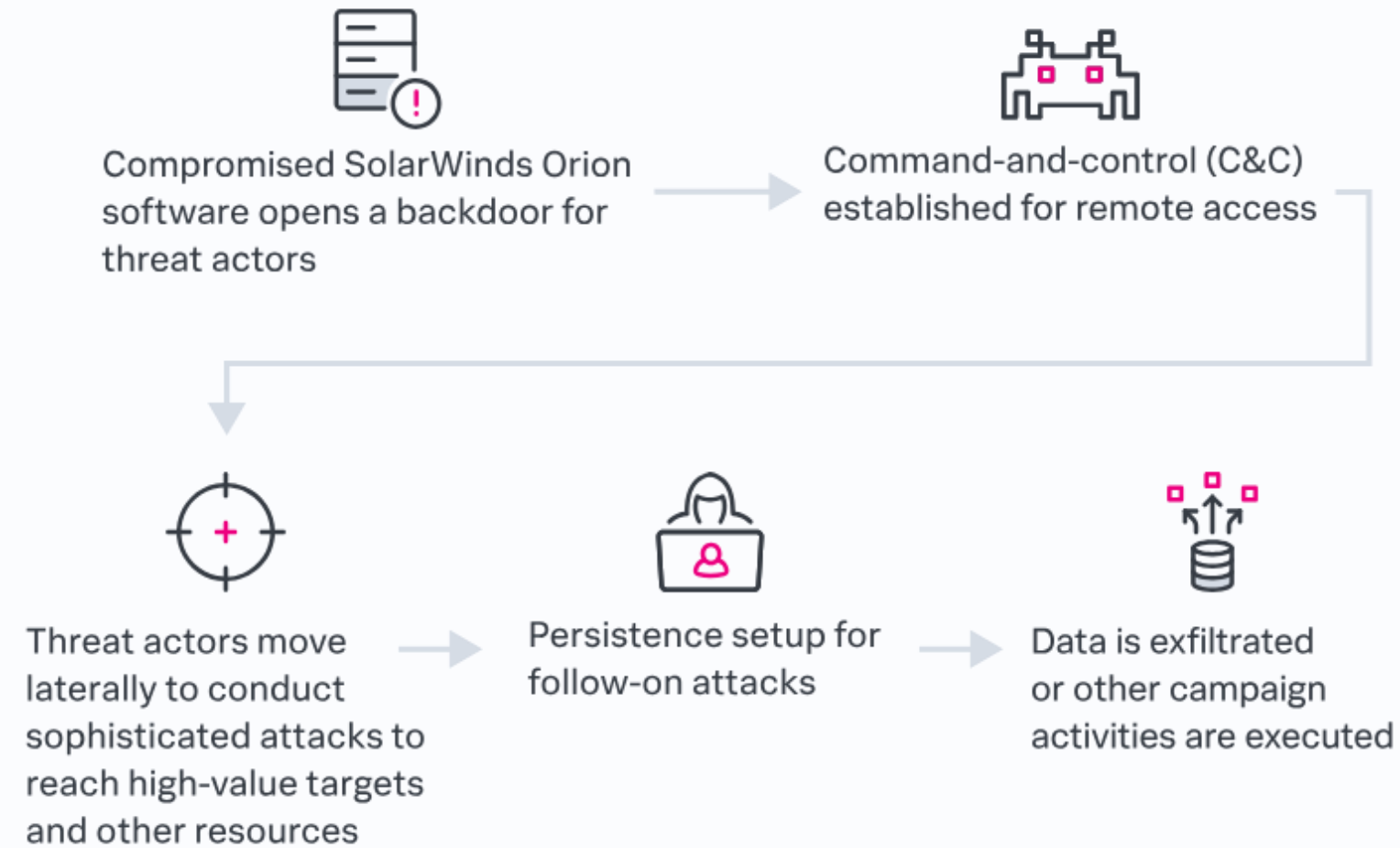
Case Study: SolarWinds Attack

Attackers infiltrated SolarWinds' software development environment, injecting malicious code into the Orion platform used by **over 18,000 organizations**. The malware update was digitally signed and distributed through legitimate channels.

AI's Role: Attackers used machine learning to evade detection systems, adapting their behavior based on victim environment characteristics. The malware remained dormant for weeks, learning normal network patterns before activating.

Impact: Multiple U.S. government agencies and Fortune 500 companies were compromised, with attackers maintaining access for months before discovery.

Lessons Learned: Even trusted software updates require verification. Zero-trust principles and continuous monitoring are essential.

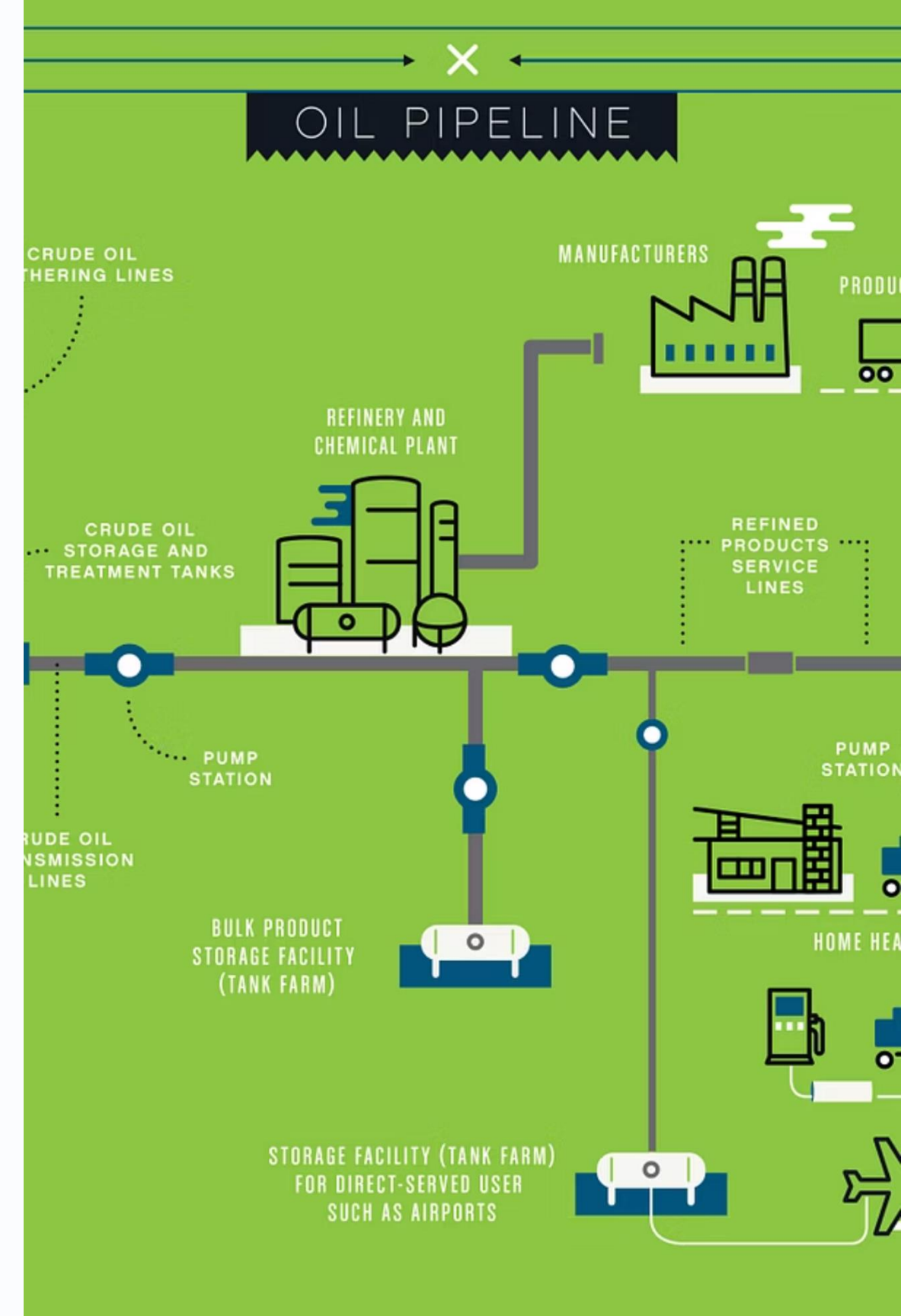


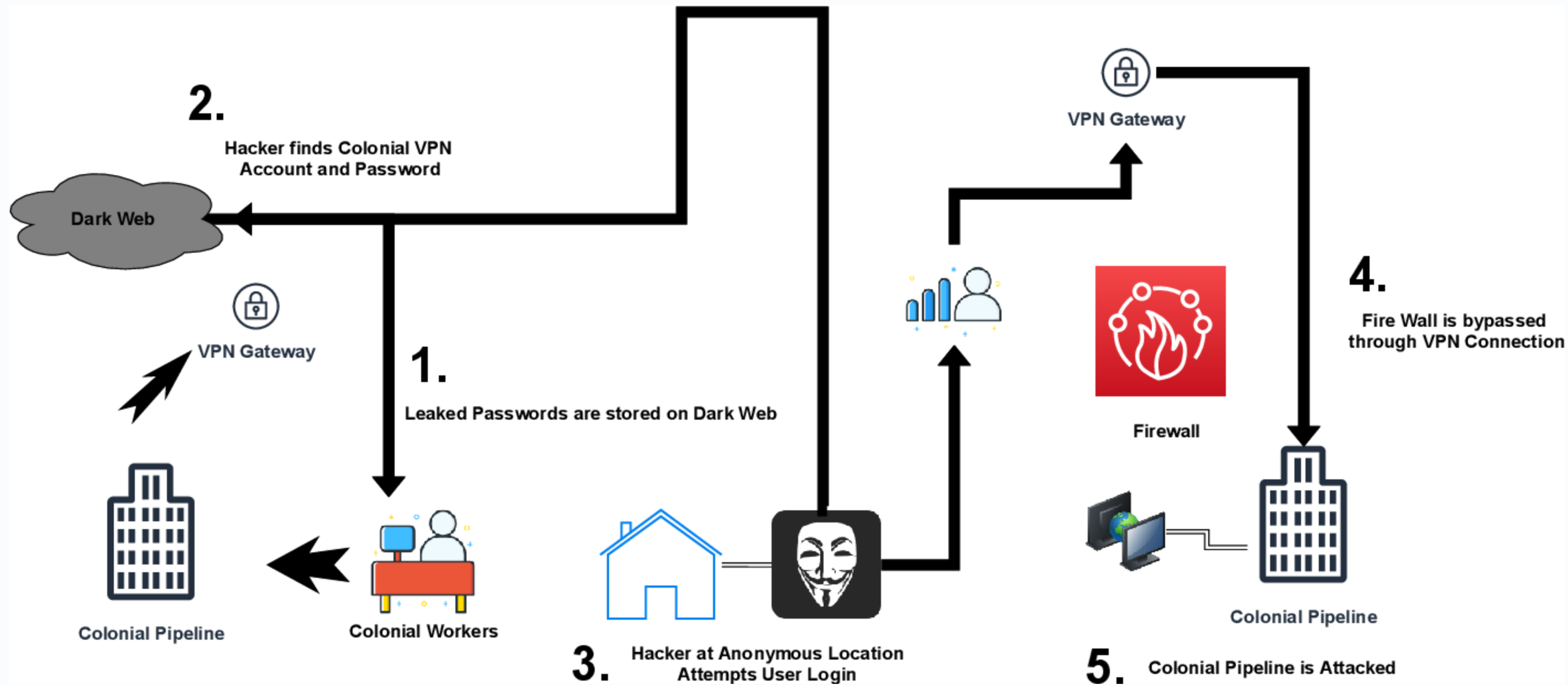
Case Study: Colonial Pipeline Attack

In May 2021, ransomware attackers crippled Colonial Pipeline, which supplies **45% of the East Coast's fuel**. The incident caused widespread panic buying, fuel shortages, and declared states of emergency across multiple states.

AI-Driven Target Selection: Attackers used machine learning to prioritize targets based on ransomware payment likelihood, company financial health, and critical infrastructure dependence. The algorithms identified Colonial Pipeline as a high-value target with significant pressure to pay quickly.

The company **paid \$4.4 million in ransom** (partially recovered by the FBI).





Spot the Fake AI Video

https://www.bilibili.com/video/BV1QokrBCEmN/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

The Double-Edged Sword

AI represents both our greatest security challenge and most powerful defense.

The same technologies that enable sophisticated attacks also provide unprecedented protection capabilities.

Success requires understanding both sides of this equation and making informed decisions about AI deployment and defense.



Key Takeaways

AI Transforms the Threat Landscape

Attackers leverage machine learning for evasion, data poisoning, and automated scale that traditional defenses cannot match.

Zero-Trust is Essential

Traditional security fails in the AI era. Zero-trust architecture, assuming breach and requiring continuous verification, provides necessary protection layers.

Defense Requires AI-Powered Tools

Manual security processes cannot keep pace with AI-driven attacks. Organizations must adopt AI-powered detection, response, and prediction to remain secure.

Continuous Learning is Critical

AI security evolves rapidly. Professionals must stay current through training and hands-on experience with emerging threats and defensive technologies.

References

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- DevSecOps AI - <https://devsecopsai.today/ai-in-cybersecurity-stop-90-of-cyber-attacks-before-they-even-start-ba545936cba5>
- UK Government: Cyber Security Risks to AI - <https://www.gov.uk/government/publications/research-on-the-cyber-security-of-ai/cyber-security-risks-to-artificial-intelligence>
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Questions?

Course Feedback

